

CBSE Test Paper 02
Chapter 6 Work Energy & Power

1. The scalar product of vectors is **1**
 - a. non commutative and not distributive
 - b. non commutative and distributive
 - c. commutative and not distributive
 - d. commutative and distributive
 2. Power is defined as **1**
 - a. ability to do work
 - b. energy due to motion
 - c. the time rate at which work is done or energy is transferred
 - d. stored energy
 3. A weight of 20kg falls from a height of 10 m. The work done by the gravitational force is (Take $g = 10 \text{ m s}^{-2}$) **1**
 - a. 3000 J
 - b. 2500 J
 - c. 1500 J
 - d. 2000 J
 4. mass and energy are equivalent and are related by **1**
 - a. $E = mc^3$
 - b. $E = 2mc^2$
 - c. $E = mc^2/2$
 - d. $E = mc^2$
 5. S.I unit of power is **1**
 - a. Joule
 - b. Newton
 - c. Angstrom
 - d. Watt
 6. Why is electrical power required at all when the elevator is descending? Why should there be a limit on the number of passengers in this case? **1**
 7. Which physical terms remain conserved in an inelastic collision? **1**
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8. What is the loss in kinetic energy after a collision, if the target body is initially at rest? **1**
9. Write the advantages and disadvantages of platinum resistance thermometer. **2**
10. A bolt of mass 0.3 kg falls from the ceiling of an elevator moving down with a uniform speed of 7 ms^{-1} . It hits the floor of the elevator (length of the elevator = 3 m) and does not rebound. What is the heat produced by the impact? **2**
11. On complete combustion a litre of petrol gives off heat equivalent to $3 \times 10^7 \text{ J}$. In a test drive a car weighing 1200 kg including the mass of driver, runs 15 km per litre while moving with a uniform speed on a straight track. Assuming that friction offered by the road surface and air to be uniform, calculate the force of friction acting on the car during the test drive, if the efficiency of the car engine were = 0.5. **2**
12. Develop a relation between the co-efficient of linear expansion, co-efficient superficial expansion and coefficient of cubical expansion of a solid. **3**
13. An adult weighing 600 N raises the centre of gravity of his body by 0.25 m while taking each step of 1 m length in jogging. If he jogs for 6 km, calculate the energy utilized by him in jogging assuming that there is no energy loss due to friction of ground and air. Assuming that the body of the adult is capable of converting 10% of energy intake in the form of food, calculate the energy equivalents of food that would be required to compensate energy utilised for jogging. **3**
14. The blades of a windmill sweep out a circle of area A. **3**
- If the wind flows at a velocity v perpendicular to the circle, what is the mass of the air passing through it in time t ?
 - What is the kinetic energy of the air?
 - Assume that the windmill converts 25% of the wind's energy into electrical energy, and that $A = 30 \text{ m}^2$, $v = 36 \text{ km/h}$ and the density of the air is 1.2 kgm^{-3} . What is the electrical power produced?
15. A body of mass M at rest is struck by a moving body of mass m . Prove that fraction of the initial KE of the mass m transferred to the struck body is $4mM/(m + M)^2$ in an elastic collision. **5**
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